

TOPIC 11

FORECASTING FINANCIAL REQUIREMENTS

Learning Objectives

- Describe the purpose for financial forecasting.
- Develop a pro forma income statement to forecast a new venture's profitability.
- Determine a company's asset and financing requirements based on a pro forma balance sheet.
- Forecast a firm's cash flows.
- Provide some suggestions for effective financial forecasting

Forecasts

Forecasts

- The analysis of a firm's historical financial statements are followed by the preparation of forecasts.
- Forecasts are predictions of a firm's future sales, expenses, income, and capital expenditures.
 - ❖ A firm's forecasts provide the basis for its pro forma financial statements.
 - ❖ A well-developed set of pro forma financial statements helps a firm create accurate budgets, build financial plans, and manage its finances in a proactive rather than a reactive manner.

Forecasts

- Sales Forecast

- A sales forecast is a projection of a firm's sales for a specified period (such as a year).
- It is the first forecast developed and is the basis for most of the other forecasts.
 - ❖ A sales forecast for a new firm is based on a good-faith estimate of sales and on industry averages or the experiences of similar start-ups.
 - ❖ A sales forecast for an existing firm is based on (1) its record of past sales, (2) its current production capacity and product demand, and (3) any factors that will affect its future product capacity and product demand.

Forecasts

- Forecast of Costs of Sales and Other Items
 - Once a firm has completed its sales forecast, it must forecast its cost of sales (or cost of goods sold) and the other items on its income statement.
 - The most common way to do this is to use the percentage-of-sales method, which is a method for expressing each expense item as a percentage of sales.
 - ❖ If a firm determines that it can use the percent-of-sales method and it follows the procedures described in the textbook, then the net result is that each expense item on its income statement will grow at the same rate as sales (with the exception of items that can be individually forecast, such as depreciation).

Looking Ahead: Financial Forecasting

- Pro Forma Financial Statements

- Statements that project a firm's financial performance and condition

- Purposes of pro forma statements:

- ❖ How profitable can the firm be expected to be, given the projected sales levels and the expected sales expense relationships?
 - ❖ What will determine the amount and type of financing (debt or equity) to be used?
 - ❖ Will the firm have adequate cash flows? If so, how will they be used; if not, where will the additional cash come from?

What If You Are Starting a New Business?

Pro Forma Financial Statements

- Are used to project what a firm's financial condition will be in the future.
- Are budgetary estimates of:
 - How much money is needed
 - What expenses will be at different sales levels
 - How much profit can be expected
- Should reflect realistic assumptions about the business and future business conditions.

Pro Forma Financial Statements

- Pro Forma Financial Statements

- A firm's pro forma financial statements are similar to its historical financial statements except that they look forward rather than track the past.
- The preparation of pro forma financial statements helps a firm rethink its strategies and make adjustments if necessary.
- The preparation of pro forma financials is also necessary if a firm is seeking funding or financing.

Preparing Financial Budgets

- Budget
 - One of the most powerful tools the entrepreneur can use in planning financial operations.
- Operating Budget
 - A statement of estimated income and expenses over a specified period of time.
- Cash Budget
 - A statement of estimated cash receipts and expenditures over a specified period of time.
- Capital Budget
 - The plan for expenditures on assets with returns expected to last beyond one year.

Types of Pro Forma Financial Statements

Financial Statement	Purpose
Pro Forma Income Statement	Shows the projected financial results of the operations of a firm over a specific period.
Pro Forma Balance Sheet	Shows a projected snapshot of a company's assets, liabilities, and owner's equity at a specific point in time.
Pro Forma Statement of Cash flows	Shows the projected flow of cash into and out of a company for a specific period.

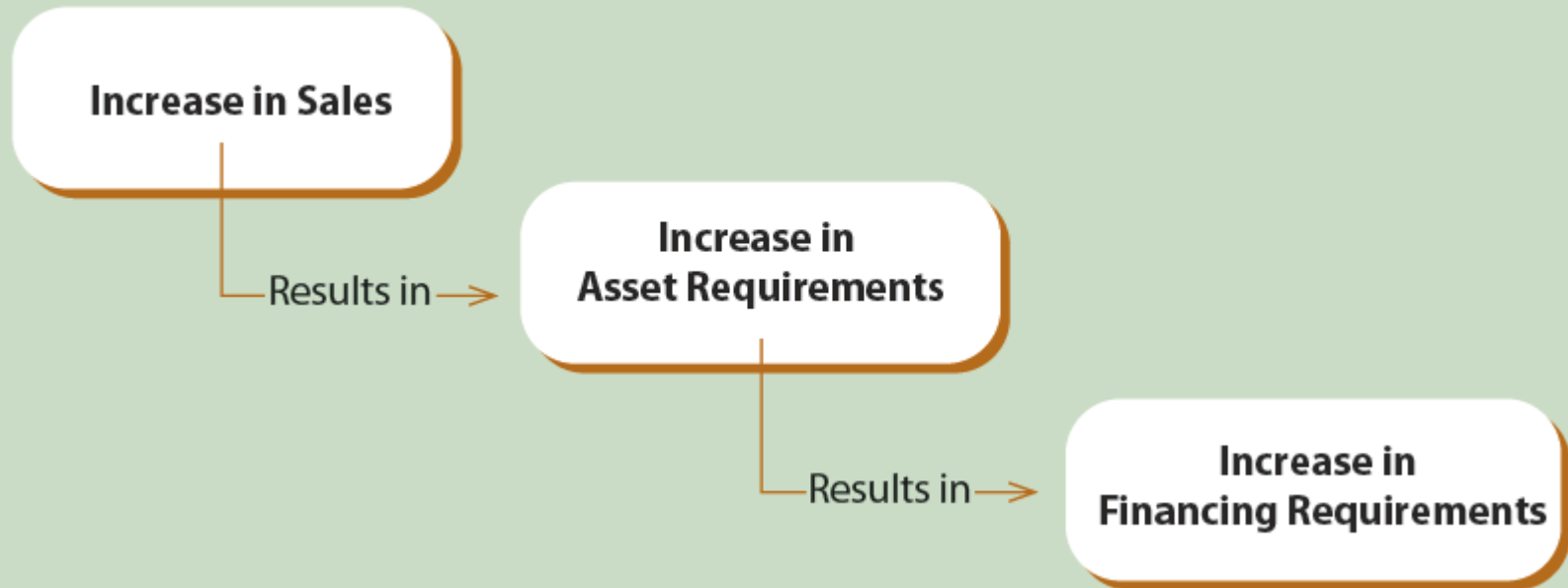
Pro Forma Income Statements for D&R Products, Inc.

	A	B	C	D	E
3	INCOME STATEMENT ASSUMPTIONS:				
4		Year 1	Year 2		
5	<u>Product sales and installations:</u>				
6	Number of projected jobs	20	30		
7	Average selling price per job	\$ 12,500	\$ 12,500		
8	Fixed cost of goods sold	\$ 100,000	\$ 100,000		
9	Fixed operating expenses	\$ 46,000	\$ 46,000		
10	Depreciation expense	\$ 4,000	\$ 4,000		
11	Interest expense	\$ 8,000	\$ 12,000		
12	Variable cost of goods sold	20%	20%		
13	Variable operating expenses	30%	30%	Equations based on assumptions	
14	<u>Product design:</u>				
15	Projected design revenues	\$ 10,000	\$ 25,000		
16	Fixed design costs	\$ 10,000	\$ 10,000		
17					
18	Income tax rate	25%	25%	Equations for:	
19					
20				Year 1	Year 2
21	<u>Sales:</u>				
22	Product sales and installations	\$250,000	\$375,000	=B6*B7	=C6*C7
23	Product design	10,000	25,000	=B15	=C15
24	Total sales	\$260,000	\$400,000	=SUM(B22:B23)	=SUM(C22:C23)
25					
26	<u>Cost of goods sold:</u>				
27	<u>Cost of goods sold: product sales and installations</u>				
28	Fixed cost of goods sold	\$100,000	\$100,000	=B8	=C8
29	Variable cost of goods sold (20% of product sales)	50,000	75,000	=B22*B12	=C22*C12
30	Total cost of goods sold: product sales and installations	\$150,000	\$175,000	=SUM(B28:B29)	=SUM(C28:C29)
31	Total cost of goods sold: product design	10,000	10,000	=B16	=C16
32	Total cost of goods sold	\$160,000	\$185,000	=SUM(B30:B31)	=SUM(C30:C31)
33	Gross profits	\$100,000	\$215,000	=B24-B32	=C24-C32
34					
35	<u>Operating expenses: product sales and installations</u>				
36	Fixed operating expenses	\$ 46,000	\$ 46,000	=B9	=C9
37	Variable operating expenses (30% of product sales)	75,000	112,500	=B13*B22	=C13*C22
38	Depreciation expense	4,000	4,000	=B10	=C10
39	Total operating expenses: product sales and installations	\$125,000	\$162,500	=SUM(B36:B38)	=SUM(C36:C38)
40	Operating profits	\$ (25,000)	\$ 52,500	=B33-B39	=C33-C39
41	Interest expense (interest rate 12%)	8,000	12,000	=B11	=C11
42	Profits before taxes	\$ (33,000)	\$ 40,500	=B40-B41	=C40-C41
43	Taxes (25% of profits before tax)	0	10,125	0	=C42*C18
44	Net profits	\$ (33,000)	\$ 30,375	=B42-B43	=C42-C43

Forecasting Asset and Financing Requirements

- Working Capital
 - Current assets, accounts receivable, and inventory required in day-to-day operations
- Net Working Capital
 - Current assets less current liabilities
- Bootstrapping
 - Minimizing a firm's investments

Assets-to-Sales Financing Relationships



Forecasting Asset and Financing Requirements (cont'd)

- Determining Asset Requirements

- Industry ratios for assets-to-sales

- ❖ Using an industry average of a “typical” firm’s assets to its sales to forecast a particular firm’s asset requirements.

- Percentage-of-sales technique

- ❖ Using a percentage of the total sales for a firm as the basis for forecasting the level of assets. accounts receivable, and inventories to be held by a firm.

Percentage-of-Sales Technique Example

Assets	Assumptions	Year 1	Year 2
Cash	4% of sales	\$ 10,000	\$ 15,000
Accounts receivable	10% of sales	25,000	37,500
Inventory	25% of sales	<u>62,500</u>	<u>93,750</u>
Total current assets		\$ 97,500	\$146,250
Gross fixed assets	Equipment and building costs	\$ 50,000	\$50,000
Accumulated depreciation	\$4,000 annually	<u>(4,000)</u>	<u>(8,000)</u>
Net fixed assets		<u>\$ 46,000</u>	<u>\$42,000</u>
TOTAL ASSETS		<u>\$143,500</u>	<u>\$188,250</u>

Determining Financing Requirements

- Basic Principles for Financing of Firms
 1. The more assets a firm needs, the greater the firm's financial requirements.
 2. A firm should finance its growth in such a way as to maintain proper liquidity.
 3. The amount of total debt used in financing a firm is limited by the funds provided by the owners.
 4. Some types of short-term debt maintain a relatively constant relationship with sales.
 5. Equity ownership comes the investments of owners, and retained earnings (profits).

Determining Financing Requirements

- Liquidity

- The degree to which a firm has working capital available to meet maturing debt obligations.

- Current Ratio

- The firm's relative liquidity, determined by dividing current assets by current liabilities.

- Debt Ratio

- Debt as a fraction of assets; total debt divided by total assets.
 - ❖ Spontaneous financing—debts such as accounts payable that increase as the firm grows.

Sources of Equity Capital

- External Equity
 - The owners' original investment
- Profit Retention
 - The reinvestment of profits in a firm
- Internal Equity
 - Capital from retaining profits within the firm
- Forecasting financial requirements (in total):

$$\begin{array}{ccccccc} \text{Total asset} & & & & & & \\ \text{requirements} & = & \text{Total sources} & = & \text{Spontaneous} & + & \text{Profits} \\ & & \text{of financing} & & \text{financing} & & \text{retained} \\ & & & & & & \text{within} \\ & & & & & & \text{business} \\ & & & & & + & \text{External} \\ & & & & & & \text{sources of} \\ & & & & & & \text{financing} \end{array}$$

Sources of Financing for D&R Products

Sources of Financing	Assumptions	Year 1	Year 2
Accounts payable	8% of sales	\$20,000	\$ 30,000
Accrued expenses	4% of sales	\$10,000	\$15,000
Mortgage	\$30,000 – \$3,000 annual payments	\$27,000	\$24,000
Common stock	Owner's investment	\$110,000	\$110,000

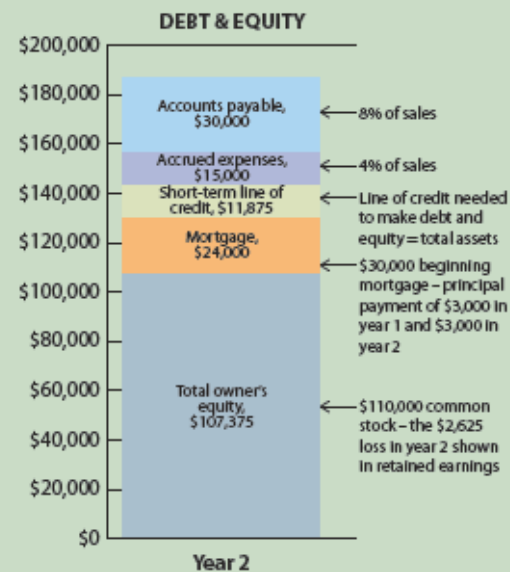
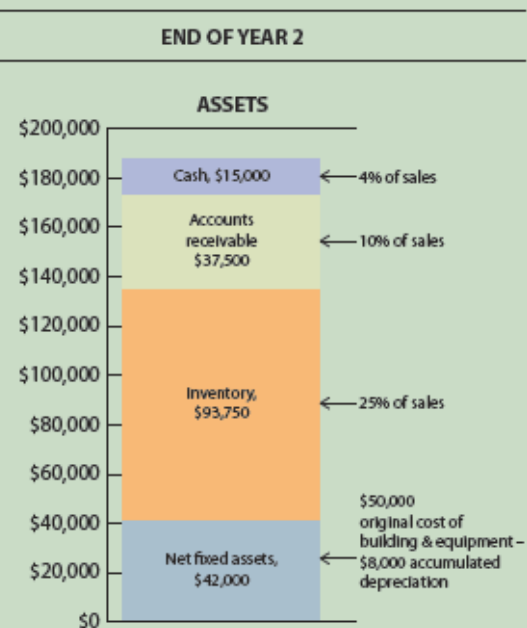
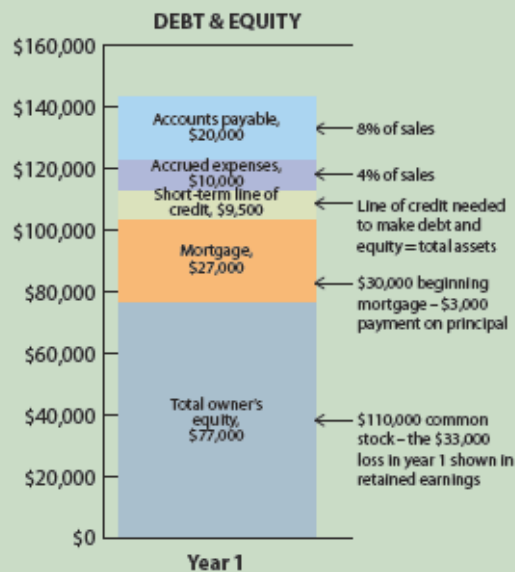
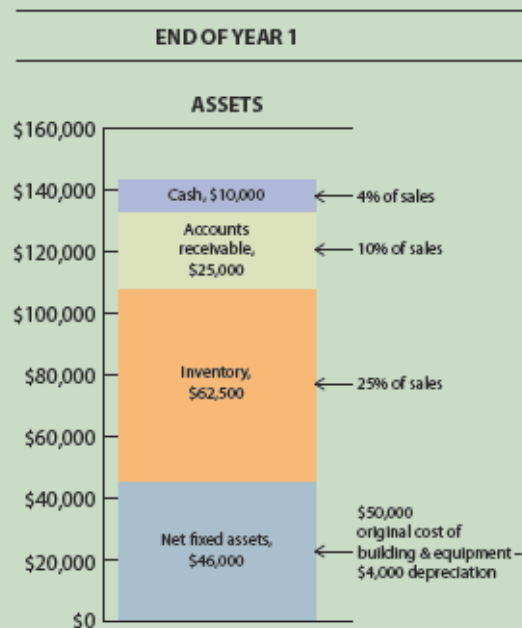
Financial Ratio Estimates for D&R Products

Ratio	Computation	Year 1	Year 2
Current ratio	$= \frac{\text{current assets}}{\text{current liabilities}}$	$= \frac{\$97,500}{\$39,500} = 2.47$	$\frac{\$146,250}{\$56,875} = 2.47$
Debt ratio	$= \frac{\text{total debt}}{\text{total assets}}$	$= \frac{\$66,500}{\$143,500} = 0.46 = 46\%$	$\frac{\$80,375}{\$188,250} = 0.43 = 43\%$

Pro Forma Balance Sheets for D&R Products, Inc.

	A	B	C	D	E
3	BALANCE SHEET ASSUMPTIONS	Year 1	Year 2		
4	Projected revenues: product sales and installations	\$250,000	\$375,000		
5	Cash/sales	4%	4%		
6	Accounts receivable/sales	10%	10%		
7	Inventory/sales	25%	25%		
8	Gross fixed assets	\$ 50,000	\$ 50,000	Equations based on assumptions	
9	Accounts payable/sales	8%	8%		
10	Accrued expenses/sales	4%	4%		
11	Cost of equipment	\$ 10,000	\$ 10,000		
12	Building cost	\$ 40,000	\$ 40,000		
13				<i>Equations for:</i>	
14	Assets			Year 1	Year 2
15	Cash	\$ 10,000	\$ 15,000	=B4*B5	=C4*C5
16	Accounts receivable	25,000	37,500	=B4*B6	=C4*C6
17	Inventory	62,500	93,750	=B4*B7	=C4*C7
18	Total current assets	\$ 97,500	\$146,250	=SUM(B15:B17)	=SUM(C15:C17)
19	Gross fixed assets	\$ 50,000	\$ 50,000	=B8	=C8
20	Accumulated depreciation	(4,000)	(8,000)	Depreciation expense for year 1	Accumulated depreciation expense for years 1 and 2
21	Net fixed assets	\$ 46,000	\$ 42,000	=B19+B20	=C19+C20
22	TOTAL ASSETS	\$143,500	\$188,250	=B18+B21	=C18+C21
23					
24	Debt (Liabilities) and Equity				
25	Accounts payable	\$ 20,000	\$ 30,000	=B4*B9	=C4*C9
26	Accrued expenses	10,000	15,000	=B4*B10	=C4*C10
27	Short-term line of credit	9,500	11,875	Required financing	Required financing
28	Total current liabilities	\$ 39,500	\$ 56,875	=SUM(B25:B27)	=SUM(C25:C27)
29	Mortgage	27,000	24,000	Original loan of \$30,000 – annual payment of \$3,000	Year 1 balance of \$27,000 – annual payment of \$3,000
30	Total debt	\$ 66,500	\$ 80,875	=SUM(B28:B29)	=SUM(C28:C29)
31	Ownership equity				
32	Common stock	\$110,000	\$110,000	Given	Given
33	Retained earnings	(33,000)	(2,625)	Year 1 loss	Year 1 loss + year 2 profit
34	Total ownership equity	\$ 77,000	\$107,375	=SUM(B32:B33)	=SUM(C32:C33)
35	TOTAL DEBT AND EQUITY	\$143,500	\$188,250	=SUM(B30:B33)	=SUM(C30:C33)
36					
37	Current ratio	\$ 2.47	\$ 2.57	=B18/B28	=C18/C28
38	Debt ratio	46%	43%	=B30/B35	=C30/C35

Pro Forma Balance Sheets for D&R Products, Inc.



Pro Forma Cash Flow Statements for D&R Products, Inc.

	Year 1	Year 2	Sources of Information
Operating activities:			
Net profits	(\$ 33,000)	\$ 30,375	← Pro forma income statement
Depreciation	4,000	4,000	
Increase in accounts receivable (cash outflow)	(\$ 25,000)	(\$ 12,500)	
Increase in inventory (cash outflow)	(\$62,500)	(\$31,250)	
Increase in accounts payable	<u>20,000</u>	<u>10,000</u>	
Cash payments for inventory	(\$ 42,500)	(\$ 21,250)	
Increase in accrued expenses	<u>10,000</u>	<u>5,000</u>	
Cash flows from operations	(\$ 86,500)	\$ 5,625	
Investing activities:			
Increase in gross fixed assets (cash outflow)	(\$ 50,000)	\$ 0	
Cash flows from investing	(\$ 50,000)	\$ 0	
Financing activities:			
Increase in short-term line of credit	\$ 9,500	\$ 2,375	
Increase (decrease) in mortgage	27,000	(3,000)	
Increase in stock	<u>110,000</u>	<u>0</u>	
Cash flows from financing	\$146,500	\$ 625	
Increase (decrease) in cash	\$ 10,000	\$ 5,000	
Beginning cash	\$ 0	\$ 10,000	
Ending cash (as shown in the balance sheets)	<u>\$ 10,000</u>	<u>\$15,000</u>	

Changes in projected balance sheets from founding of business to year 1 and from year 1 to year 2

Three-Month Cash Budget for D&R Product, Inc.

Assumptions:

Anticipated sales collections:

In the month of sale	40%
1 month later	30%
2 months later	30%

*For example, January sales of \$4,000 are collected as follows:

(40%) \$1,600 in January,
(30%) \$1,200 in February,
(30%) \$1,200 in March.

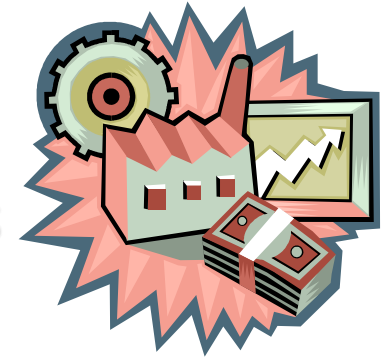
	December	January	February	March
Monthly sales	\$ 0	\$4,000	\$6,000	\$9,000
Inventory purchases on credit	\$2,400	\$3,600	\$5,400	\$7,800

	December	January	February	March
Monthly sales	\$0	*\$ 4,000	\$ 6,000	\$ 9,000
Cash receipts				
Step 1: Collection of sales				
In month of sale		\$ 1,600	\$ 2,400	\$ 3,600
1 month later			\$ 1,200	\$ 1,800
2 months later				1,200
Total cash receipts		\$ 1,600	\$ 3,600	\$ 6,600
Step 2: Cash disbursements				
Step 2a: Payments on inventory purchases		\$ 2,400	\$ 3,600	\$ 5,400
Advertising		3,000	3,000	3,000
Step 2b: Wages and salaries		5,000	6,000	6,000
Utilities		150	200	200
Total cash disbursements		\$10,550	\$12,800	\$14,600
Step 3: Cash flows from operations		(\$8,950)	(\$9,200)	(\$8,000)
Step 4: Allen's personal investment		110,000		
Step 5: Purchase of equipment and building		(50,000)		
Step 6: Mortgage (loan from the bank to buy the building)		30,000		
Step 7: Beginning cash balance		0	81,050	71,850
Step 8: Ending cash balance		\$81,050	\$71,850	\$63,850

Use Good Judgment When Forecasting

• Practical Suggestions

1. Develop realistic sales projections.
2. Build projections from clear assumptions about marketing and pricing plans.
3. Do not use unrealistic profit margins.
4. Don't limit your projections to an income statement.
5. Provide monthly data for the upcoming year and annual data for succeeding years.
6. Avoid providing too much financial information.
7. Be certain that the numbers reconcile—and not by simply plugging in a figure.
8. Follow the plan.



Forecasting Profitability

- Net Income Depends On:

- Amount of sales
- Cost of goods sold
- Operating expenses
- Interest expense
- Taxes

“If we’re doing so well, then why am I always so broke?”